

TE 456: Satellite Communication and Navigation System

Legacy Satellite Communication to NTN
What & Why



Course Information

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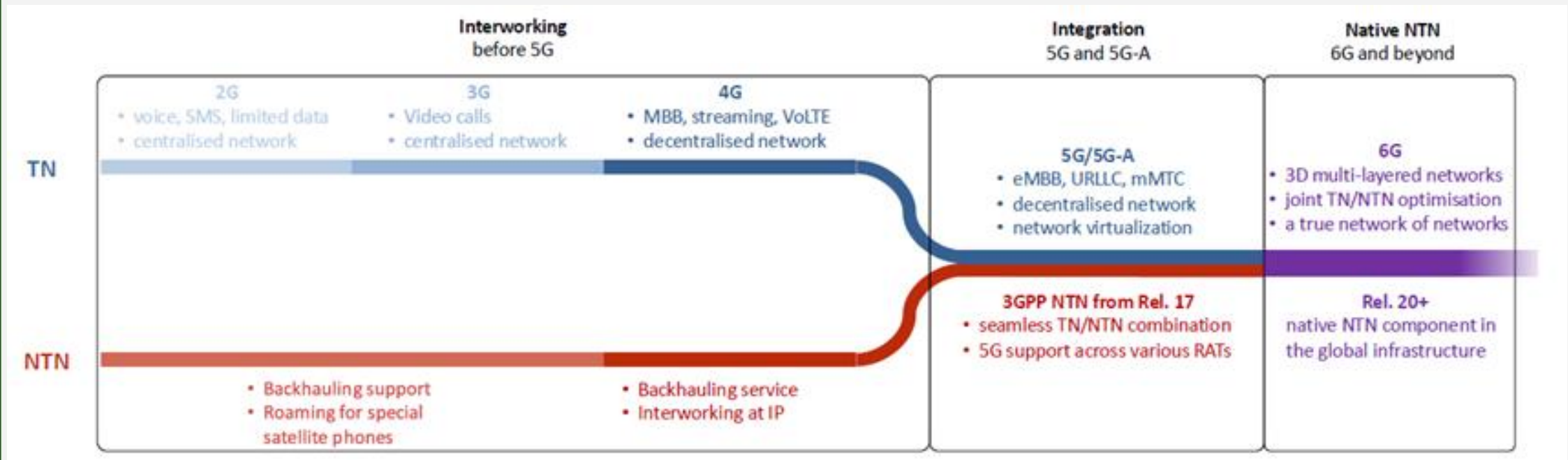
Introduction

- The content of TE456 is
 - focused on **legacy satellite communication**
 - Where satellite connectivity required **proprietary hardware** and isolated networks for **specific missions**
 - Communication, environmental monitoring, navigation, surveillance
- For a considerable period
 - Satellite and terrestrial mobile communication industries have developed as **two separate** industrial chains
- Currently
 - The ecosystem of satellite and terrestrial mobile communication industries has been progressively converging



Introduction

■ TN – NTN Integration/Unification



Introduction

- Telecom Ecosystem
 - The 3rd Generation Partnership Project (3GPP)
 - completed the first global 5th generation (5G) New Radio (NR) standard
 - paving the way for making 5G a commercial reality
- Due to
 - Technological advancements and demand for Global/ubiquitous connectivity services
 - There is a resurgence of interest in providing connectivity from space/air
- Focus is now on
 - Enabling 5G NR to support satellite communications
 - to further expand the 5G/6G ecosystems



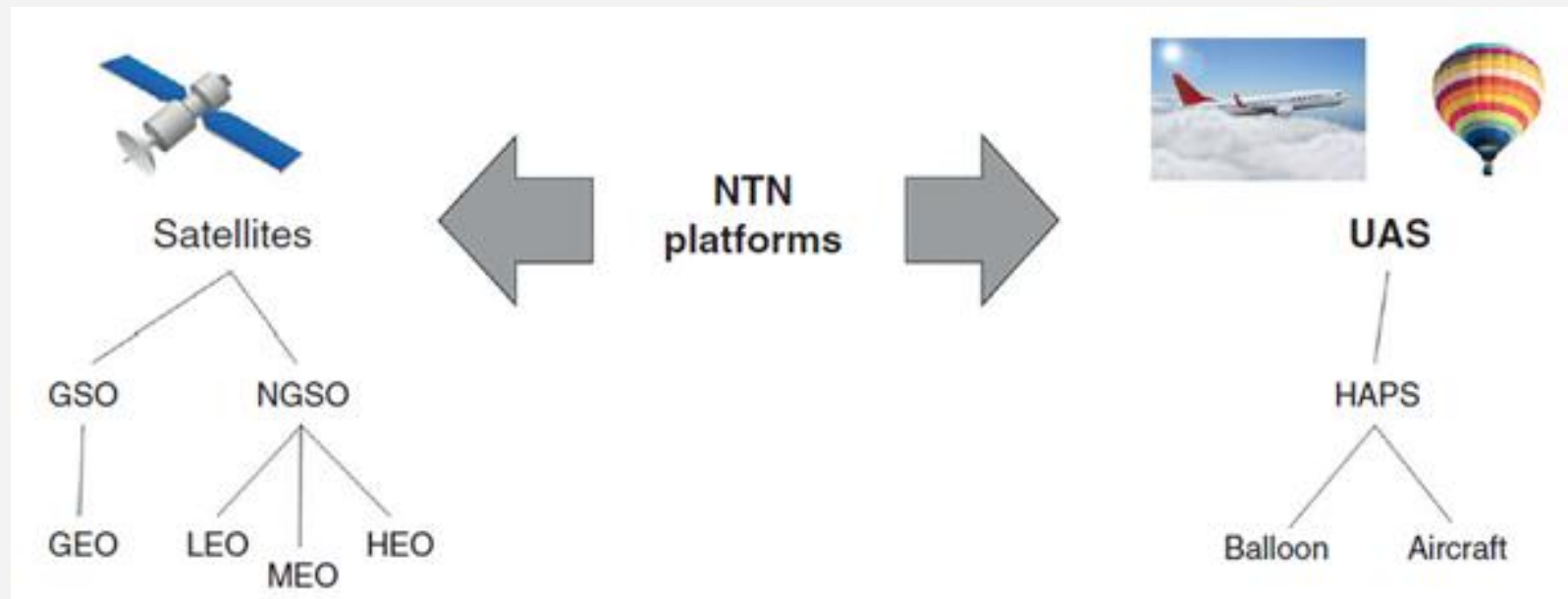
Introduction

- Emerging
 - Focus is now on integrating spaceborne and airborne (NTN) systems
 - directly into standard 3GPP (5G/6G) mobile (TN) ecosystems to provide **global connectivity**
 - TN = Terrestrial Network
 - NTN = Non-Terrestrial Network
- It is anticipated that
 - The integration of satellite communications into 5G will facilitate
 - **anything, anytime, anywhere** connectivity in the 5G era and beyond
- We will change the focus of the course
 - From legacy Satellite Communication to Non-Terrestrial Networks



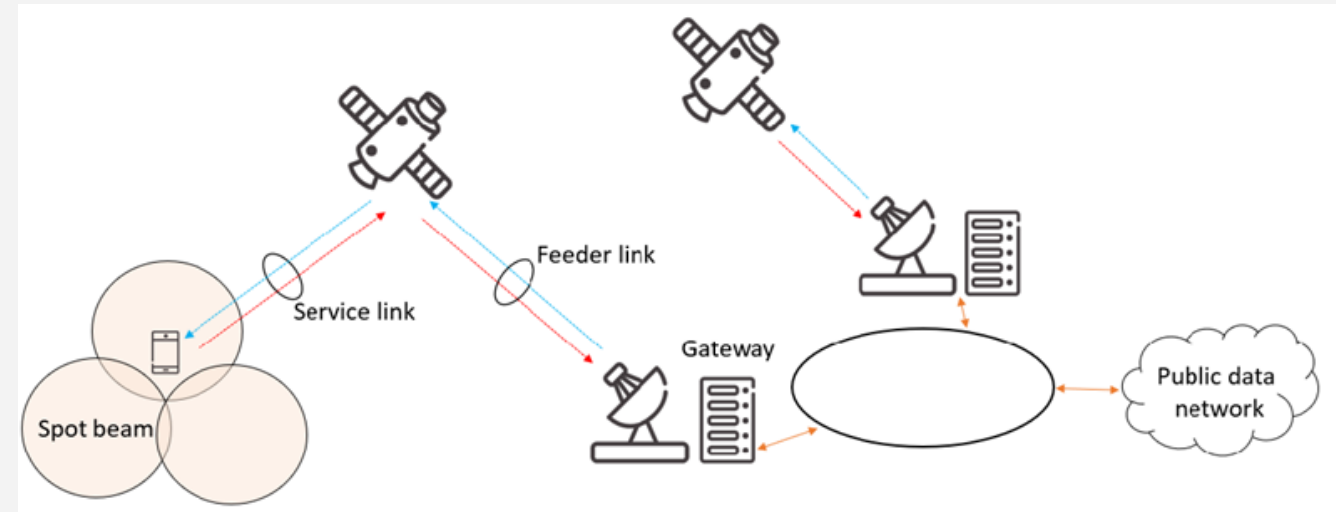
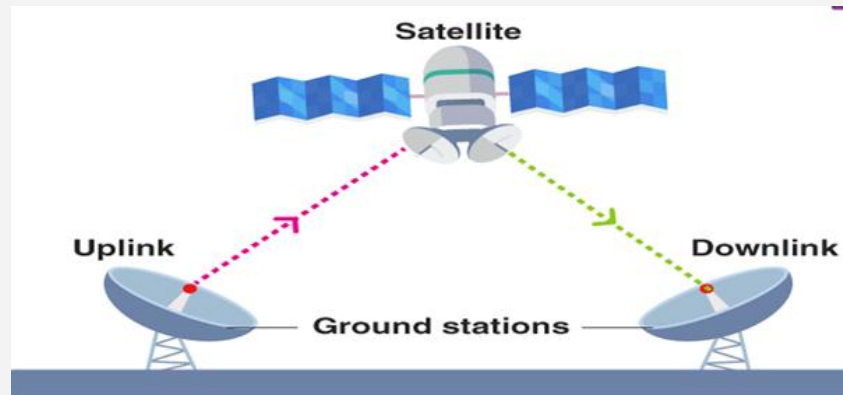
Introduction

- Spaceborne and Airborne vehicles



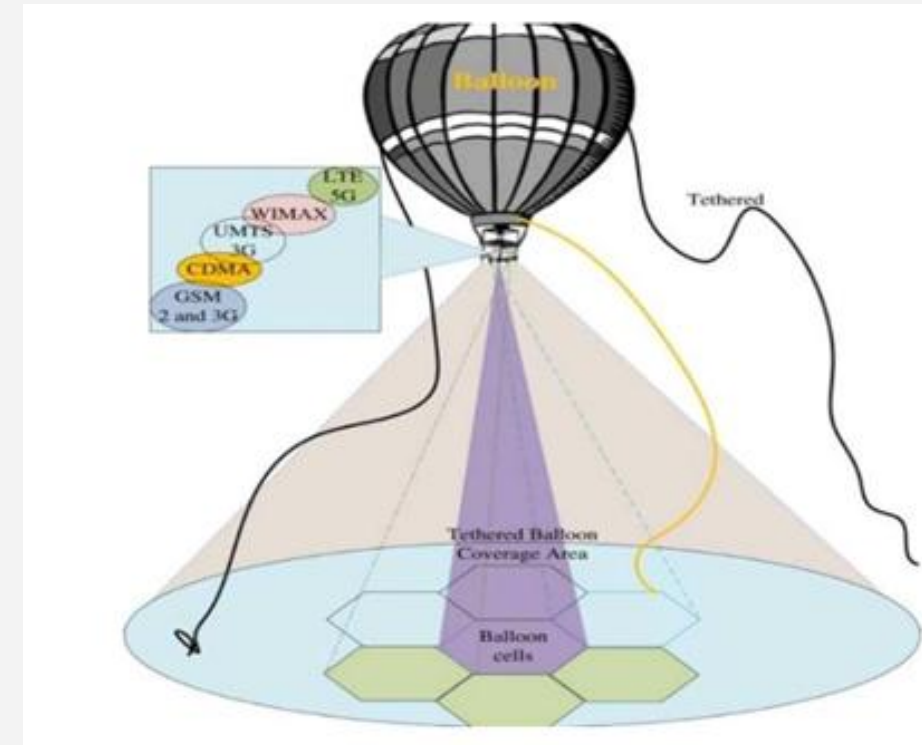
Introduction

- Spaceborne = satellite Communication
 - can be defined as the use of artificial satellites to establish communication links between various points on the Earth's surface
 - i.e., a telecommunication system encompassing at least one communication satellite



Introduction

- Airborne
 - Also, there is significant
 - advancements related to Airborne systems (drones, and High-Altitude Platform Stations (HAPS)),
 - initially to support the connectivity to on-ground users in critical scenarios, but now also for many other applications



Introduction

- The combination of
 - Spaceborne and Airborne systems led to a broadening in the family of communications from the sky – Non-Terrestrial Networks
- Non-Terrestrial Networks (NTN)
 - is a wireless communication network that utilizes **radio network equipment** placed on an **airborne** or **spaceborne** vehicle



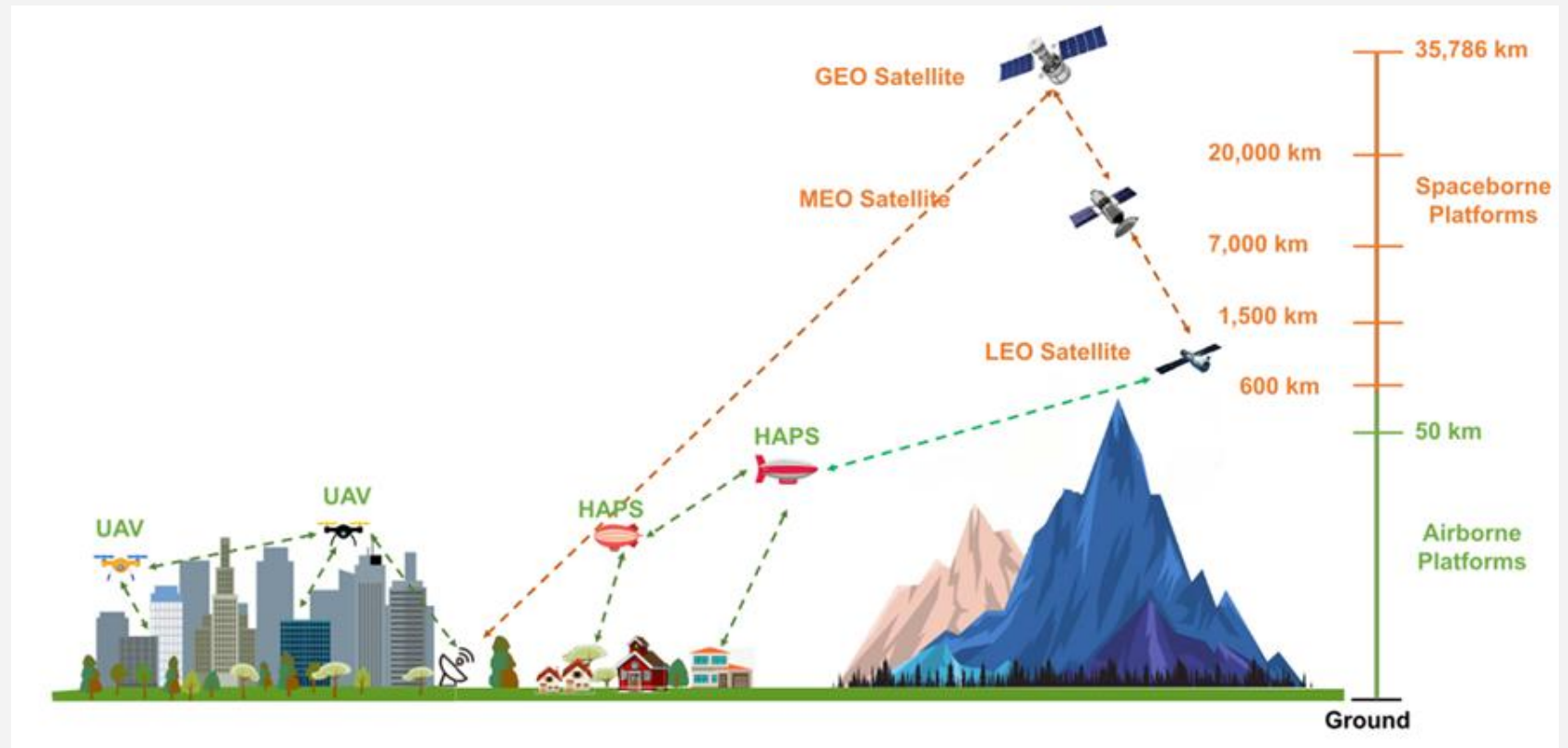
NTN – What

- Non-Terrestrial Networks (NTNs)
 - are wireless communication systems that utilize airborne or spaceborne platforms—such as satellites (LEO, MEO, GEO), High-Altitude Platform Stations (HAPS), and drones (UAVs)
 - rather than solely relying on ground-based cell towers.
 - They extend 5G and future 6G connectivity to remote areas, oceans, and skies, providing seamless, global coverage where terrestrial infrastructure is impossible or impractical

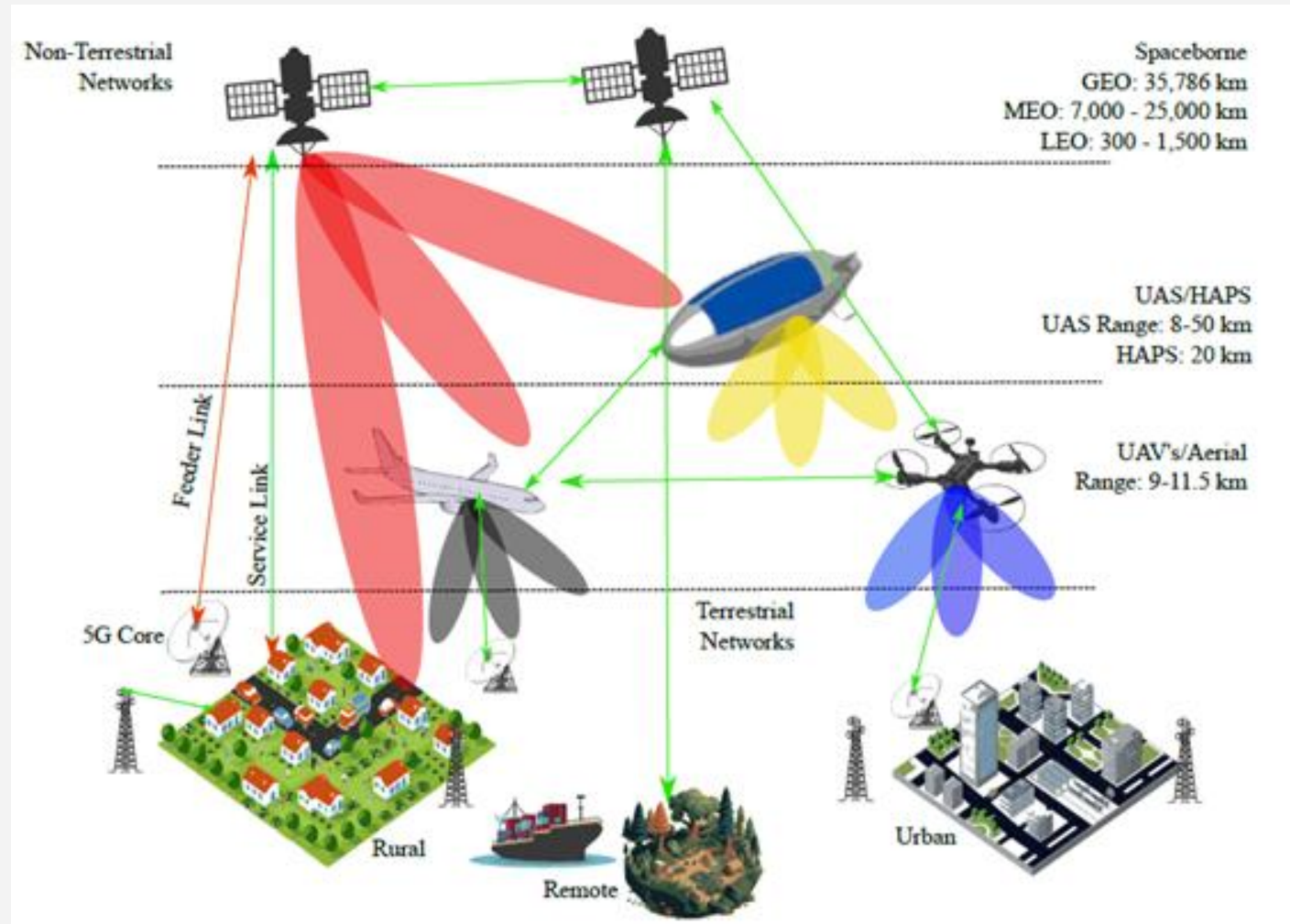


NTN – What

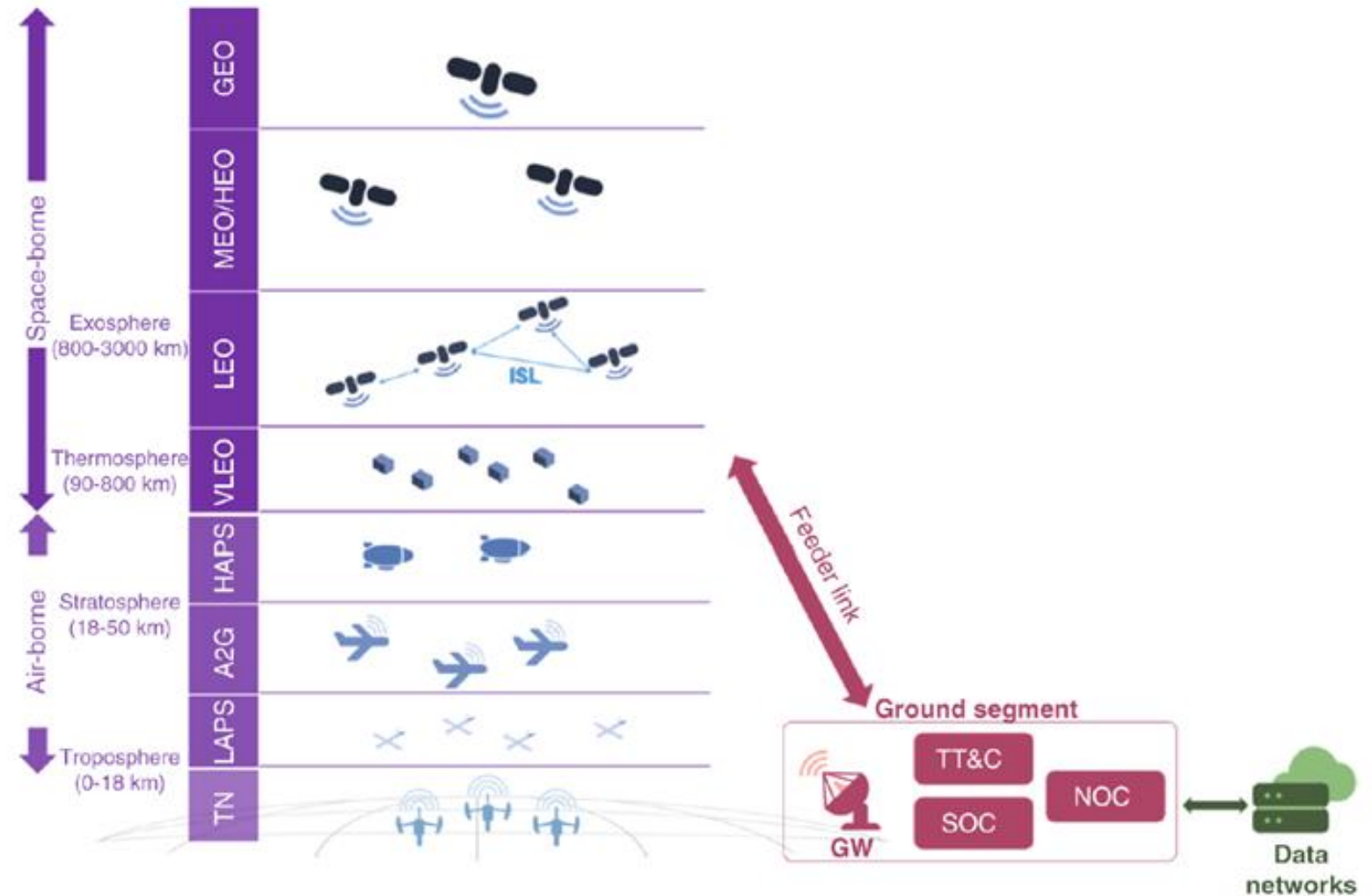
- NTN:
 - A network that utilizes a communications **platform** at the altitude of more than tens of kilometers
 - Platforms: Satellites (Ex: GEO, MEO, and LEO), High-Altitude Platform Station (HAPS) and UAVs



NTN – What



NTN – What



NTN- What

- Non-Terrestrial Networks (NTN)
 - are communication networks that leverage airborne (e.g., unmanned aerial vehicles) and spaceborne vehicles (e.g., satellites) to facilitate ultra-reliable communications and connectivity with high data rates and low latency over expansive regions
 - refer to wireless communication systems that use **airborne** or **spaceborne** platforms (like satellites and drones) rather than traditional ground-based **cell towers**
 - are advanced wireless communication systems that operate above the Earth's surface using satellites, high-altitude platform stations (HAPS), and drones. Unlike traditional terrestrial networks that rely on ground-based infrastructure



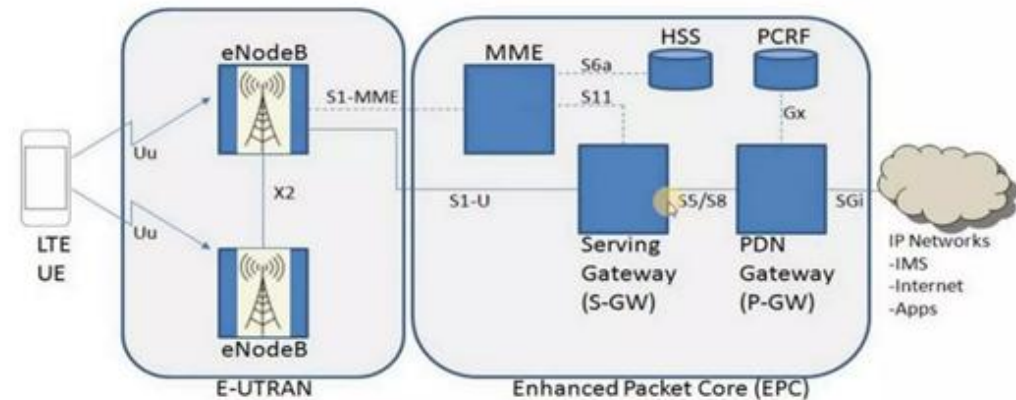
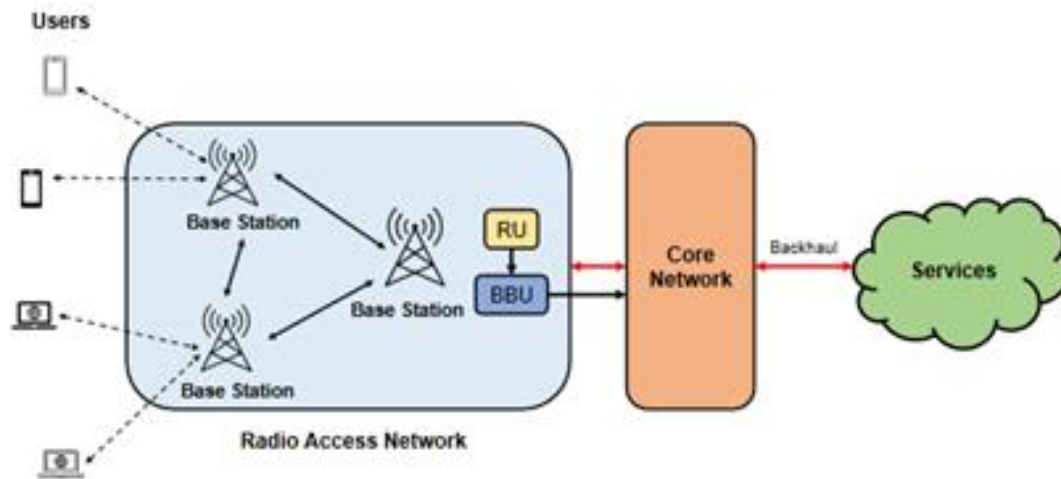
NTN - Why

- A TN can provide
 - good quality of service (QoS) to users when the users are in the radio coverage of the TN
- Service providers
 - carry out coverage and capacity planning to determine the number and locations of base stations in a target service area to meet the expected traffic demands
- These base stations need
 - to connect to other base stations to support features such as handover through a transport network.
 - to connect to the core network via a transport network, because the core network interfaces with data networks such as the Internet



NTN - Why

- Terrestrial networks
 - rely on ground-based infrastructure like cell towers and fiber cables for localized, high-speed coverage



NTN - Why

- Cost-effective deployment
 - of the base stations and the associated transport network is not feasible in areas with sparse populations and remote and vast areas on land, such as mountains, deserts, and national parks.
- In cases such as
 - large portions of oceans and remote islands, deployment of a terrestrial cellular network is challenging due to inadequate land infrastructure and the lack of a business case



NTN - Why

- 6G Vision

- Global connectivity

- Fully connected world

- Provide ubiquitous wireless connectivity for ALL

- Terrestrial networks

- i. cover less than 40% of the Earth's surface
 - ii. highly susceptible to disruption from environmental hazards and natural disasters



5G

- Africa
 - Approximately 1.2% of total mobile connections on the continent
- Global
 - global average of over 20%



TN - Ghana

- 3G all district capitals
- 4G currently accounts
 - for slightly over 43% of mobile internet penetration in Ghana
- 4G networks
 - primarily concentrated in urbanized economic hubs like Accra and Kumasi
 - rural regions have lagged, leaving a large portion of the unconnected demographic dependent on 3G
- Next-Gen InfraCo (NGIC) rollout
 - This neutral infrastructure company is tasked with rolling out 3,200 4G sites and 1,200 5G sites across the country



TN - Ghana

■ 5G

- The government has set a highly ambitious target of reaching 70% population coverage by March 2027 to boost broadband access
 - Infrastructure deployments are actively underway, with nearly 50 active sites installed across regions including Greater Accra, Ashanti (including Kumasi)



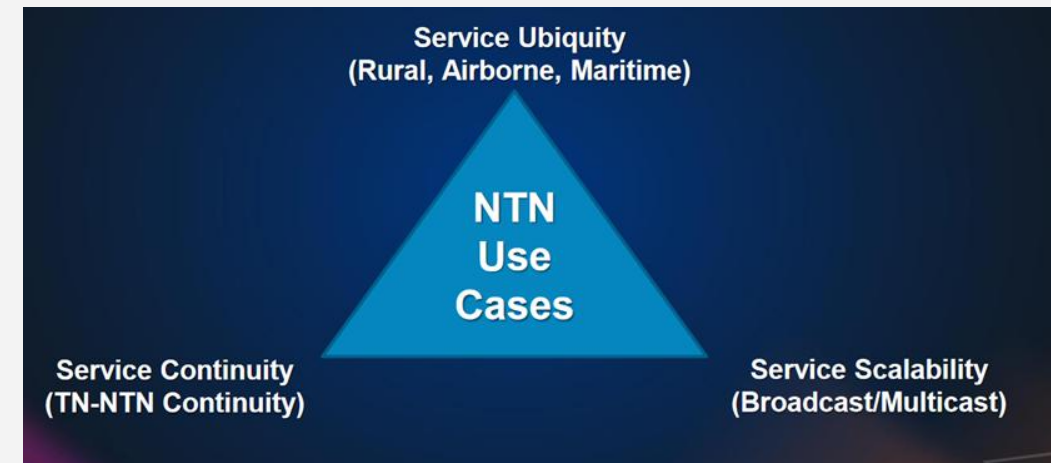
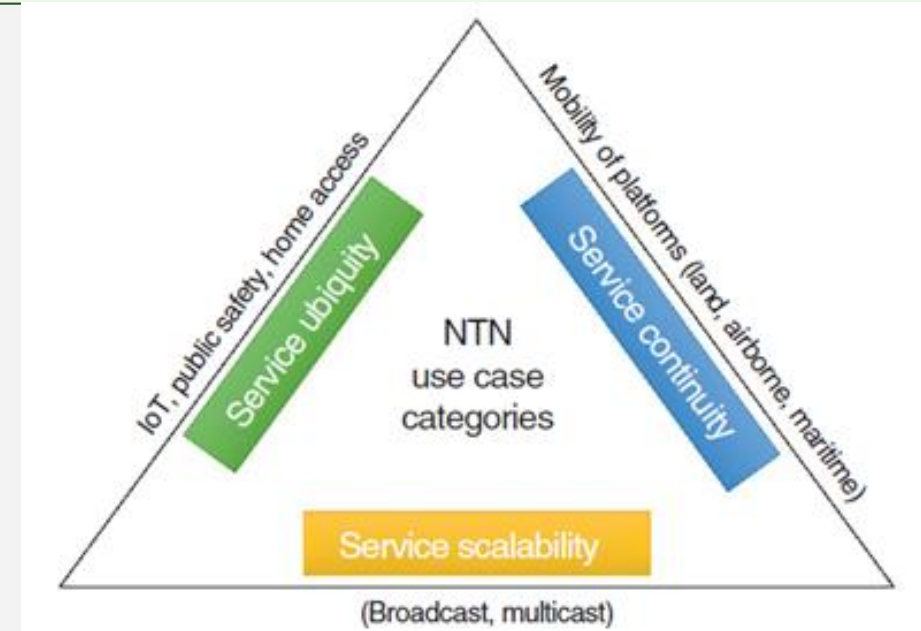
5G rollout

- Non-Terrestrial Networks
 - are expected to foster the rollout of 5G service in
 - unserved areas that cannot be covered by terrestrial 5G network (isolated/remote areas, on board aircrafts or vessels) and
 - underserved areas (e.g. sub-urban/rural areas) to upgrade the performance of limited terrestrial networks in cost effective manner



NTN - Why

- An NTN
 - offers an attractive solution in the challenging environments mentioned
- Three NTN use case categories
 - represent one of the key motivations behind the design and deployment of an NTN
 - because TNs cannot adequately address these use cases



NTN - Why

- **Service ubiquity**

- An NTN platform can create large beams to provide radio coverage anywhere on the Earth
 - Remote land areas, homes, and IoT devices (e.g., devices in smart farms and manufacturing plants) in underserved or unserved rural areas with sparse populations, vast mountains, and oceans can all be illuminated by beams from spaceborne vehicles
- Furthermore, the infrastructure of TNs may be damaged or destroyed due to natural disasters (e.g., earthquakes and floods) and armed conflicts such as wars
 - Through a suitable design, NTN platforms can connect to data networks on the Earth's surface via a gateway



NTN - Why

- **Service continuity**

- There are several cases in which the cellular subscriber moves out of the radio coverage of cellular networks.
 - For example, moving out of a suburban area into a rural area, boarding a cruise ship, or during the takeoff in case of air travel often results in the loss of radio coverage and services.
 - Cellular subscribers and IoT devices may be on moving land platforms (e.g., cars, trains, and trucks), airborne platforms (e.g., commercial or private aircraft), and maritime platforms (e.g., cruise ships and container or cargo ships).
- An NTN can provide
 - service continuity in such cases by providing radio coverage in the areas that are not covered by a TN



NTN - Why

▪ Service scalability

- Cellular systems aim for high efficiency to optimize the network performance and the user experience while meeting resource constraints such as the available amount of spectrum.
 - When the same service content needs to be sent to a target set of users in a large geographic area, all the base stations in such a geographic area need to consume precious radio resources
- An NTN cell covers a much larger area compared to a TN cell, resulting in increased efficiency.
 - Examples of the services that can benefit from large-area broadcasts include over-the-air software updates and selected entertainment programs, such as live games and prescheduled movies and shows.



NTN - Why

- Service Ubiquity
 - With connections from the air or space
 - NTN can empower ultra-reliable communications when terrestrial infrastructures are not available, such as in remote and unreachable areas or during natural disasters
- Service scalability
 - NTN can easily provide multicast/broadcast connectivity with high data rates and low latency over a large region, enabling massive Machine Type Communication (mMTC) and enhanced Mobile Broadband (eMBB) communications
- Service continuity
 - multiple NTN can be connected together and/or connected to existing terrestrial networks to provide continuous and ubiquitous wireless coverage
 - Thus bringing us closer to the era of anything, anytime, and anywhere communications



Course Objective

- This course
 - provides students with the fundamental understanding of Non-Terrestrial Networks (NTN) that is, **Satellites**, **HAPs**, **UAVs**, and their integration with 5G/6G networks to achieve Global/universal connectivity
 - introduces students to the fundamental concepts and techniques for the design and analysis of satellite communication/Non-Terrestrial Networks and navigation systems, and the integration of Terrestrial Network (TN) and Non-Terrestrial Networks



Course Outline

- Ground, space, user segments
- user and feeder links
- orbits and constellations
- Antennas, beams, and payload architectures
- Technical Challenges
- Mobility, handover, timing, and synchronization in NTN
- link budget



Course Outline

- 5G NR architecture and NTN integration
- NB-IoT and direct-to-device NTN
- Positioning principles
- Positioning principles applied to satellite systems
- The GPS system
- The Galileo system



Activities/Labs

- **Lab 1:** Orbit and coverage visualization
 - Seeing NTN. Simulate GEO, MEO, and LEO satellites and compare coverage time, elevation angle, and latency.
- **Lab 2:** Link budget
 - Can the signal reach? Compute received power, link margin, etc. for different orbit types.
- **Lab 3:** Doppler shift in LEO NTN
 - Compute Doppler shift at different carrier frequencies and satellite speeds.
- **Lab 4:** Propagation delay comparison
 - Compare communication delay for terrestrial, LEO, MEO, and GEO systems.
- **Lab 5:** NR NTN channel modeling/simulation
 - Use or adapt MathWorks' NTN channel example.
- **Lab 6:** NTN handover or satellite visibility analysis
 - Simulate when a ground station, UE should switch from one LEO satellite to another



Reading Materials:

1. Dennis Roddy (2006) Satellite Communications, 4th edition, McGraw-Hill
2. Gerard Maral, Michel Bousquet and Zhili Sun (2020) Satellite Communication Systems: Systems, Techniques and Technology, 6th Edition, Wiley
3. Vanelli-Coralli et al., 5G Non-Terrestrial Networks: Technologies, Standards, and System Design, Wiley-IEEE Press, 2024
4. E. Kaplan and C. Hegarty, Understanding GPS/GNSS: Principles and Applications, Third Edition:, ser. Engineering professional collection. Artech House Publishers, 2017





TN vs SatCom

■ TN - Success

- A major driver of the success of
 - Terrestrial mobile networks over the past few decades have been
 - the international standardization effort
 - The international standardization effort helps ensure compatibility among vendors and reduces network operation and device costs

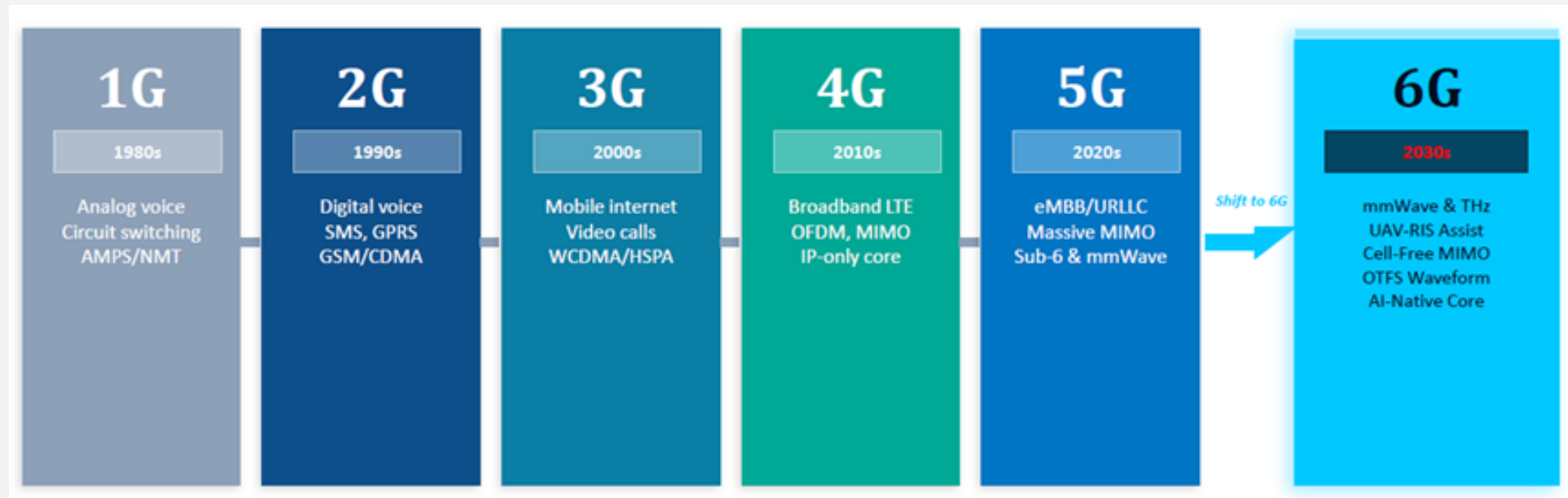
■ SatCom

- Due to lack of standardization
 - The interoperability between different satellite solution vendors has been difficult and the availability of devices is limited, leading to an overall fragmented satellite communications market up to date

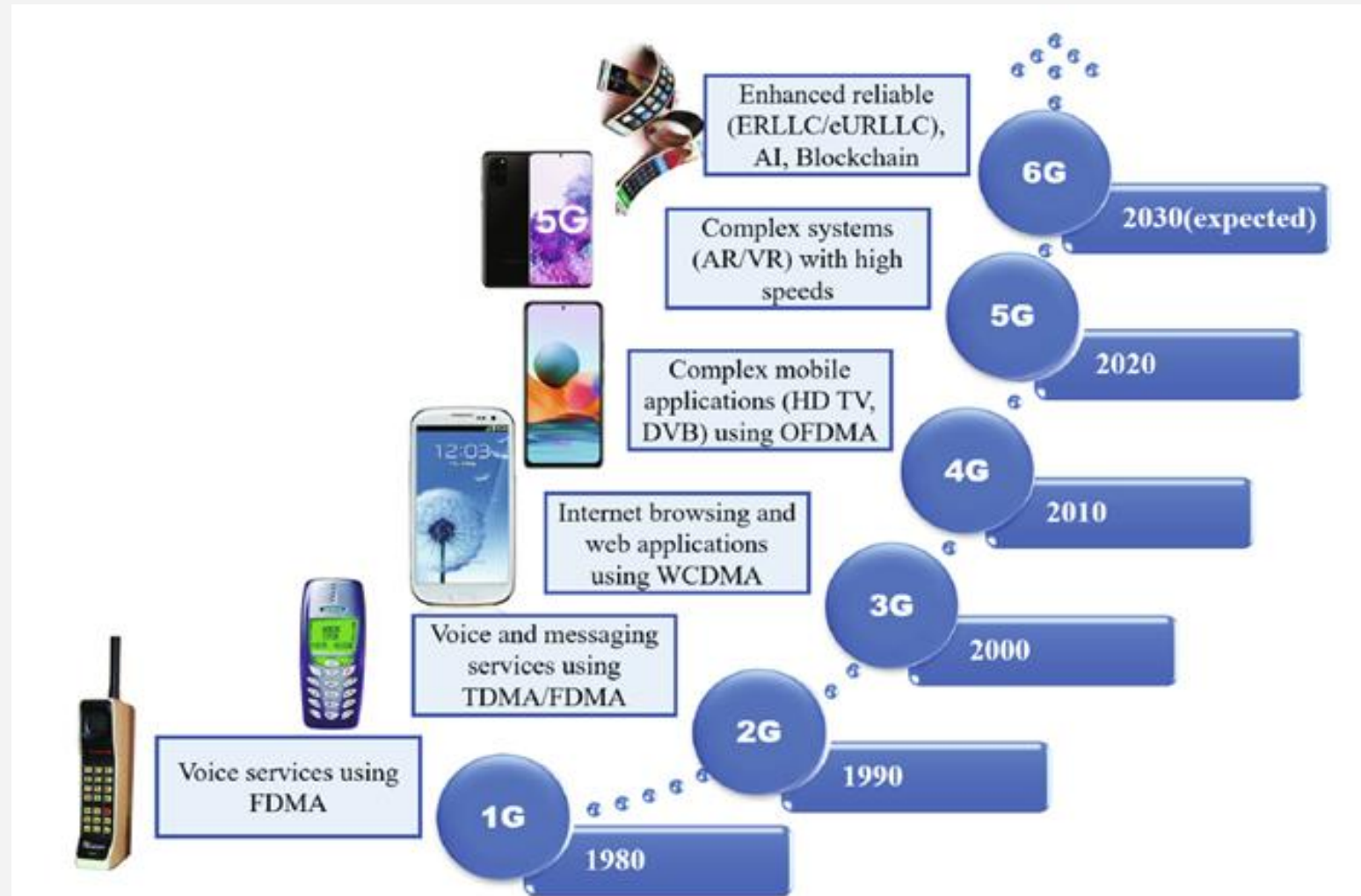


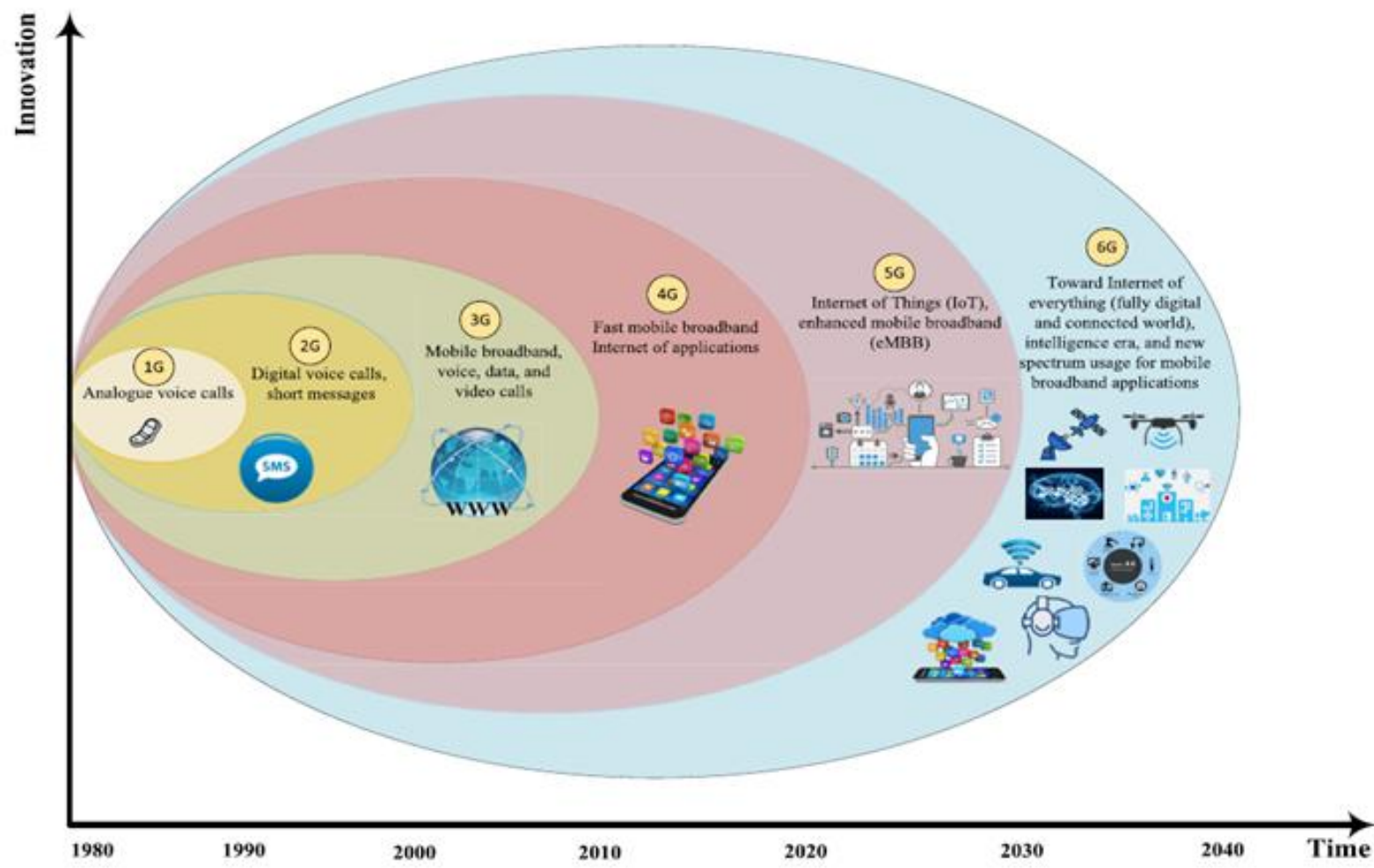
Terrestrial Networks (TN) - What

- Terrestrial networks
 - rely on **ground-based infrastructure** like cell towers and fiber cables for localized, high-speed coverage



Evolution





The evolution of mobile generation from 1G to 6G.



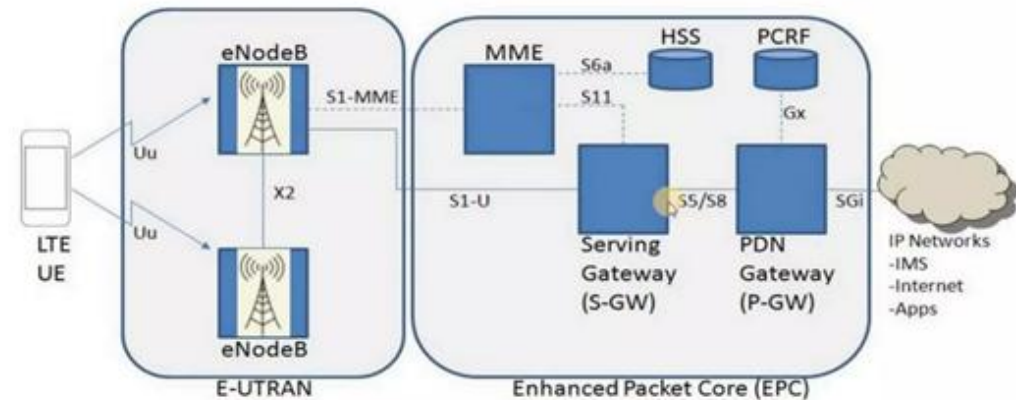
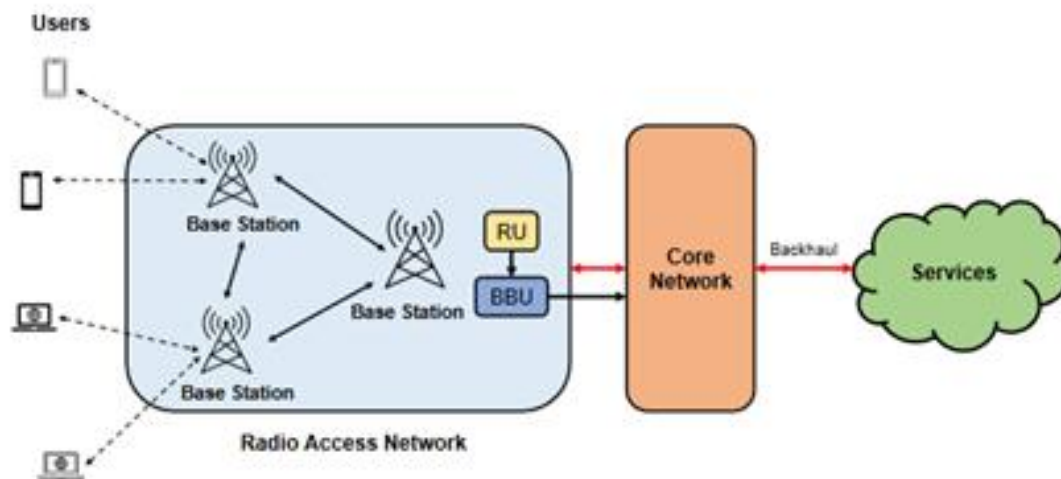
TN – Generations of Communication Technology

Generations of communication technology				
Sl. no	Year	Generation	Key features	Limitations
1	1980s	1G	• Voice signals • Analog cellular phones • NMT, AMPS	Low security
2	1990s	2G	• Text messages • Voice signal • Digital signal • GSM, TDMA, CDMA, GPRS, EDGE	Low internet support
3	1998s	3G	• Voice signals • Data signals • Video signals • Wireless and fixed internet access • W-CDMA, UMTS	Low high-speed internet support
4	2008s	4G	• Higher data rate • Interoperability protocol • Mobile IP	Does not help with the connected devices are more than 50 billion
5	2019s	5G	• Increased connectivity • IoT • Low latency • Greater transmission speed	



Terrestrial Networks (TN) - What

- Terrestrial networks
 - rely on ground-based infrastructure like cell towers and fiber cables for localized, high-speed coverage



Key Characteristics of TN

- **Infrastructure:**

- Comprised of fixed, ground-based base stations (e.g., NodeB/eNodeB/gNodeB), physical cables, and switching centers, as seen in 5G

- **Low Latency:**

- Offers low propagation delay (usually ms for data services) compared to satellite systems, which is crucial for real-time applications

- **High Reliability:**

- Generally, provides higher reliability and lower bit error rates than non-terrestrial networks (NTN).



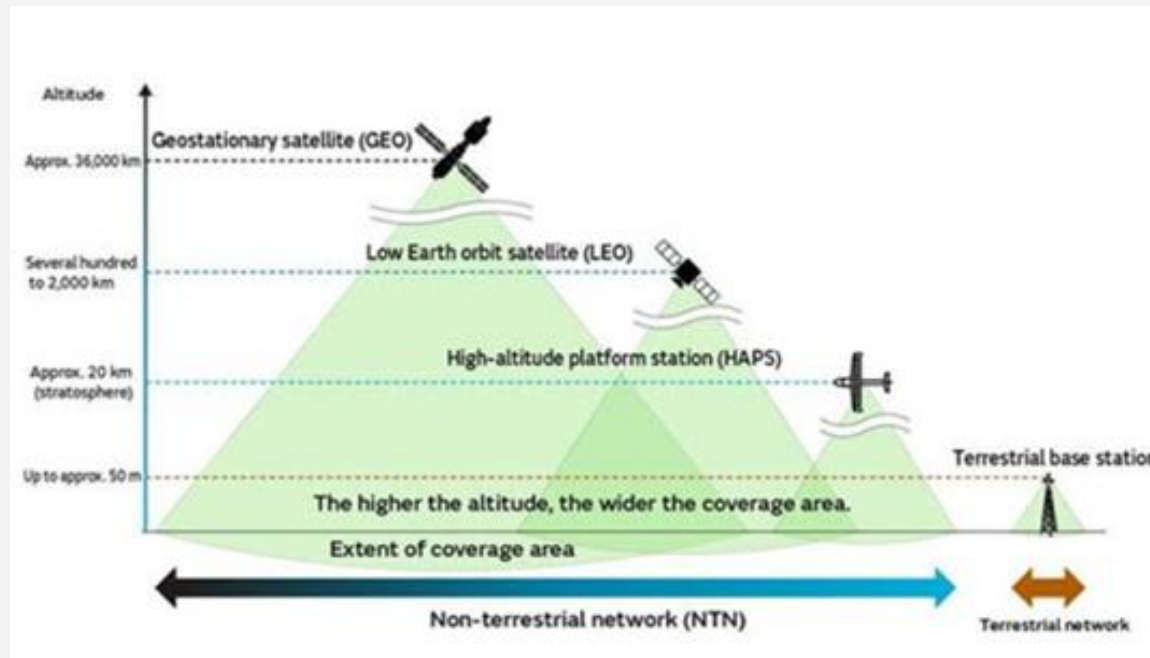
Key Characteristics of TN

- **High Bandwidth:**
 - Optimized to handle high data rates for user equipment
- **Service Types:**
 - Supports a wide range of applications, including high-speed internet, voice services, and urban IoT devices
- **Stable Coverage:**
 - Provides consistent, stable coverage areas, though they can be limited in rural or mountainous regions



TN vs NTN

- TN and NTN
 - differ primarily in their **infrastructure** and **coverage areas**



TN vs NTN

Feature	TN	NTN
Infrastructure	Ground-based cell towers, masts, and physical cables	Satellites (LEO, MEO, GEO), High-Altitude Platforms (HAPs), and drones
Coverage Area	Localized, high-density coverage (often limited by physical geography)	Global, ubiquitous coverage including oceans, deserts, and rural areas
Cell Size	Small, allowing for high bandwidth and capacity per cell	Massive, ranging from tens of kilometers to thousands of kilometers in diameter.
Latency	Ultra-low, typically < 1ms	higher, ranging from about 20ms for (LEO) up to 500ms + for (GEO) due to large distances
Signal Constraints	Standard path loss; limited atmospheric interference	severe path loss, high Doppler shifts (due to satellite mobility), and potential atmospheric/ionospheric interference
Primary Use Cases	Urban and suburban mobile broadband, IoT, and high-speed internet	Rural/maritime connectivity, disaster response, aerospace communications, and global IoT tracking.



TN vs NTN

Aspect	Non-Terrestrial Networks (NTN)	Terrestrial Networks
Coverage Area	Global (up to 3500 km)	Regional (up to 100km)
Communication Rates	Up to 150 Mbps (Starlink)	More than 100 Mbps on average
Latency	Higher latency (more than 500ms for GEO satellite)	Lower latency (lower than 40ms)
Infrastructure Cost	Very high for satellites	Lower for ground infrastructure
Reliability	Robust to disasters	Vulnerable to disasters, single points of failure
Accessibility	Can reach remote/underserved areas	Difficulty in remote area coverage
Mobility Support	Seamless mobility across wide areas	Frequent handovers for mobility



TN - Limitations

- The primary limitations of TN
 - include restricted global reach
 - leaving rural areas and oceans disconnected
 - vulnerability to physical damage during natural disasters, and high deployment costs in rugged terrain



NTN - Why

- Terrestrial networks
 - form the fundamental infrastructure of modern communication systems, serving more than 4 billion users globally
- However
 - terrestrial networks are facing a wide range of challenges,
 - from **coverage** and **reliability** to **interference** and congestion
 - As the demands of the 6G era are expected to be much higher, it is crucial to address these challenges to ensure a robust and efficient communication infrastructure for the future
- To address these problems
 - Non-terrestrial Network (NTN) has emerged to be a promising solution



NTN - Why

- The need for Non-Terrestrial Networks (NTNs)
 - arose from the **limitations** of traditional terrestrial infrastructure, which struggles to provide coverage in remote or underserved regions and can fail during natural disasters
- NTNs offer a solution by
 - extending connectivity to these areas and providing a backup in emergencies
 - addressing the growing demand for global coverage driven by technologies like IoT and 5G
- Additionally
 - NTNs can be more cost-effective than extensive terrestrial networks and serve strategic and scientific purposes
 - Advances in satellite and high-altitude platform technology have made NTNs more viable, enabling them to complement and enhance terrestrial networks



NTN- Why

- The limitations inherent in ground-based infrastructure, alongside economic considerations
 - often hinder the deployment of TNs in remote or inaccessible regions such as rural areas, deserts, and oceans
 - Consequently, user equipment (UEs) remains unable to access terrestrial services within these underserved areas
- Leveraging NTN alongside existing terrestrial infrastructure
 - emerges as a viable and cost-effective solution to ensure **uninterrupted** and **widespread** wireless coverage



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